

the strength of your belief in the occurrence of a given event), subjective to the observer, who expresses it as rationally as he wishes under some constraints. These constraints of consistency in decision making are obvious: you cannot bet there is a 60 percent chance of snow tomorrow *and* a 50 percent chance that there will be no snow. The agent needs to avoid violating something called the Dutch book constraint: that is, you cannot express your probabilities inconsistently by engaging in a series of bets that lock in a certain loss, for example, by acting as if the probabilities of separable contingencies can add up to more than 100 percent.

There is another difference here, between “true” randomness (say the equivalent of God throwing a die) and randomness that results from what I call epistemic limitations, that is, lack of knowledge. What is called ontological (or ontic) uncertainty, as opposed to epistemic, is the type of randomness where the future is not implied by the past (or not even implied by anything). It is created every minute by the complexity of our actions, which makes the uncertainty much more fundamental than the epistemic one coming from imperfections in knowledge.

It means that there is no such thing as a long run for such systems, called “nonergodic” systems—as opposed to the “ergodic” ones. In an ergodic system, the probabilities of what may happen in the long run are not impacted by events that may take place, say, next year. Someone playing roulette in the casino can become very rich, but, if he keeps playing, given that the house has an advantage, he will eventually go bust. Someone rather unskilled will eventually fail. So ergodic systems are invariant, on average, to paths, taken in the intermediate term—what researchers call absence of path dependency. A nonergodic system has no real long-term properties—it is prone to path dependency.

I believe that the distinction between epistemic and ontic uncertainty is important philosophically, but entirely irrelevant in the real world. Epistemic uncertainty is so hard to disentangle from the more fundamental one. This is the case of a “distinction without a difference” that (unlike the ones mentioned earlier) can mislead because it distracts from the real problems: practitioners make a big deal out of it instead of focusing on epistemic constraints. Recall that skepticism is costly, and should be available when needed.

There is no such thing as a “long run” in practice; what matters is what happens before the long run. The problem of using the notion of “long run,” or what mathematicians call the asymptotic property (what happens when you extend something to infinity), is that it usually makes us blind to what happens before the long run, which I will discuss later